

HW 4: Block Diagram Algebra and Feedback

The textbook exercises, designated with the word “Nise”, are from the “Problems” section at the end of the chapter in *Control System Engineering* by Norman Nise, 7th edition.

This assignment uses exercises from the textbook. The additional instructions are to clarify and/or expand each exercise.

Nise 5.1

Nise 5.11

Do this problem both manually (“pen and paper”) and with MATLAB

1. Manually, find the closed-loop transfer function using block diagram algebra. Then calculate the requested metrics using the key equations handout (see first week’s handout).
2. Using MATLAB, use the `feedback()` and `stepinfo()` commands to accomplish the same result.

Exercise 3: Proportional Speed Control of the USV

This exercise connects block diagram algebra to the feedback controllers we will implement and test in the USV field labs. For background on the surge plant model and proportional control, refer to the *PID Control: From Analysis to Practice* article distributed in class.

Background

The surge (forward-speed) dynamics of the USV can be modeled as a first-order plant:

$$G_{\text{surge}}(s) = \frac{5.1}{1.3s + 1}$$

where the input is normalized throttle (± 1 , dimensionless) and the output is forward speed (m/s). We want to regulate speed by closing a unity feedback loop with a proportional controller $C(s) = K_p$.

Tasks

1. **Block diagram.** Draw the standard unity-feedback block diagram with $G_{\text{surge}}(s)$ in the forward path and K_p as the controller block. Label the reference $R(s)$, error $E(s)$, control effort $U(s)$, and output $Y(s)$.
2. **Closed-loop transfer function.** Using the feedback form of block diagram algebra, derive the closed-loop transfer function $T(s) = Y(s)/R(s)$. Rewrite $T(s)$ in the standard first-order form and express the closed-loop DC gain K_{cl} and closed-loop time constant τ_{cl} as explicit functions of K_p .
3. **Design for rise time.** Find the minimum value of K_p such that the closed-loop rise time satisfies $T_r < 1.5$ s. Show your work: express $K_{p,\text{min}}$ analytically as a formula before evaluating numerically.
4. **MATLAB verification.** Use `feedback()` to build $T(s)$ and `stepinfo()` to extract metrics, plot the closed-loop unit-step response for K_p .

5. **Report.** State the following values obtained from your MATLAB results:

- The value of $K_{p,\min}$
- Closed-loop rise time T_r (from `stepinfo`)
- Steady-state error e_{ss}

Steady-state error is the difference between the desired output and the actual steady-state output. For a unity step reference, $e_{ss} = 1 - K_{cl}$, where K_{cl} is the DC gain of the closed-loop transfer function. Verify that your analytical value for e_{ss} is consistent with the unit step response graph produced by MATLAB.

Briefly explain why $e_{ss} > 0$ for any finite K_p with proportional-only control.

Exercise 4: Feedforward Compensation

This exercise extends Exercise 3 by adding a *feedforward* path to eliminate the steady-state error without changing the closed-loop dynamics.

Background

In Exercise 3 you showed that proportional-only control always leaves a nonzero steady-state error. One approach to eliminating e_{ss} is simply to increase K_p until the error is negligible. However, there are practical reasons why arbitrarily large K_p is undesirable:

- **Actuator saturation.** For the USV, normalized throttle is bounded to ± 1 . At the instant a unit step is applied, the error is $E(0) = 1$, so the initial control command is $U(0) = K_p$. Any $K_p > 1$ immediately saturates the throttle, violating the linear model assumption and causing behavior the transfer function cannot predict.
- **Noise amplification.** A proportional controller multiplies sensor noise directly by K_p . High gain passes measurement noise through to the actuator, causing excessive motor/thruster activity and mechanical wear.
- **Sensitivity to unmodeled dynamics.** The plant model G_{surge} captures only the dominant surge mode. Higher loop gain reduces gain and phase margins, making the closed loop more vulnerable to higher-frequency effects (hull flex, propeller dynamics, wave forces) that were ignored in the first-order model.

A better strategy is to keep K_p modest and add a *feedforward* path to handle the steady-state offset directly.

1. **Feedforward block diagram.** A feedforward gain K_{ff} is added from the reference $R(s)$ directly to the summing junction at the plant input, so the control effort becomes

$$U(s) = K_p E(s) + K_{ff} R(s).$$

Draw the modified block diagram. Label $R(s)$, $E(s)$, $U(s)$, $Y(s)$, K_p , K_{ff} , and $G_{\text{surge}}(s)$.

2. **Closed-loop transfer function with feedforward.** Derive the closed-loop transfer function $T_{ff}(s) = Y(s)/R(s)$ for the feedforward-augmented loop. Show that choosing

$$K_{ff} = \frac{1}{K_{\text{plant}}}, \quad K_{\text{plant}} = G_{\text{surge}}(0) = 5.1(m/s),$$

gives $T_{ff}(0) = 1$, i.e., zero steady-state error to a step reference, while leaving the closed-loop time constant τ_{cl} unchanged compared to the P-only design.

3. **MATLAB comparison.** Using the same K_p found in Exercise 3:
- Build the feedforward closed-loop transfer function *analytically* (express it as a `tf()` object) and verify with `dcgain()`.
 - Plot the unit-step responses of the P-only and P+feedforward controllers on the same axes.
 - Use `stepinfo()` to report the rise time and steady-state value (or confirm $e_{ss} = 0$) for the feedforward case.
4. **Report.** State the following values from your MATLAB results:
- K_{ff} (analytical value)
 - Closed-loop rise time T_r for the P+feedforward controller
 - Steady-state error e_{ss} for the P+feedforward controller

Comment on how T_r compares to the P-only result from Exercise 3 and explain why the rise time is (or is not) changed by the feedforward gain.